Operation Management

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<MBACCZG526, Operation Management>
Lecture No. 3

Production System

Input	Process	Output
Material	Cutting, Drilling	Goods
Machines	Teaching	Services
Labour	Transportation	
Capital	Farming	
Facilities	Debugging	
Energy	Analysing	
	Searching	

Design of Goods & Services

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The objective of the Goods & Service design decision is to develop and implement a product strategy that meets the demands of the marketplace with a competitive advantage.

Reason for Design of Goods & Services

- Political, liability, or legal.
- Economic
- Social and demographic
- Technological.
- Environmental.
- Competitive (SWOT)

Take Home Discussion

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Do the SWOT Analysis of Apple.

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Design of Goods / Product

Defining a Product

Most manufactured items, as well as their components, are defined by :

- Engineering drawing.
- Bill of material (BOM).

- **Make-or-buy decision** —The choice between producing a component or a service and purchasing it from an outside source.
- **Group technology** —A product and component coding system that specifies the size, shape, and type of processing; it allows similar products to be grouped.

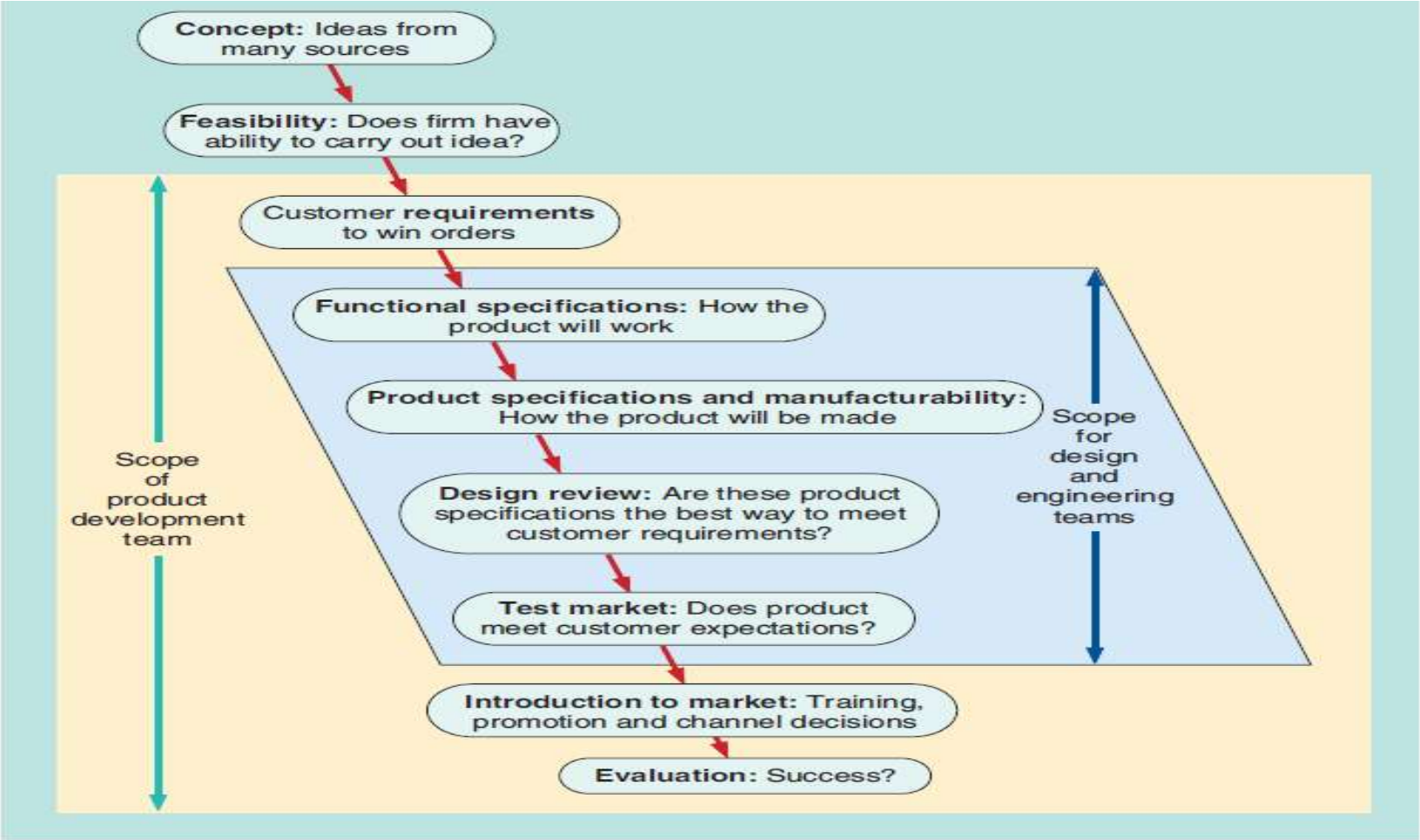
Documents for Production

- Assembly drawing.
- Assembly chart.
- Route sheet.
- Work order.
- Configuration management.
- Product life cycle management (PLM).
- Engineering change notice (ECN) : A correction or modification of an engineering drawing or bill of material.

New Product Strategy

- **Market pull :**
- **Technology push :** In this view, technology is the primary determinant of the products the firm should make.
- **Inter-functional view :** This view holds that the product should not only fit the market needs (new or existing markets) but have a technical advantage as well.

Generate New Product



Source: Heizer, Render, Muckson

Product Development

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- Quality Function Deployment & House of Quality.
- Organizing for Product Development.
- Manufacturability and Value Engineering.
- Product Lifecycle.

Source: Heizer, Render, Muckson

Quality Function Deployment

- QFD is a structured approach to define customer needs or requirements and translating them into specific plans to produce products to meet those needs.
- QFD originated in Bridgestone tires and Mitsubishi heavy industries when they used quality charts that take customer requirements into account in their product design process.
- It is an orderly process to determine the critical quality characteristics.

House of Quality (QFD)

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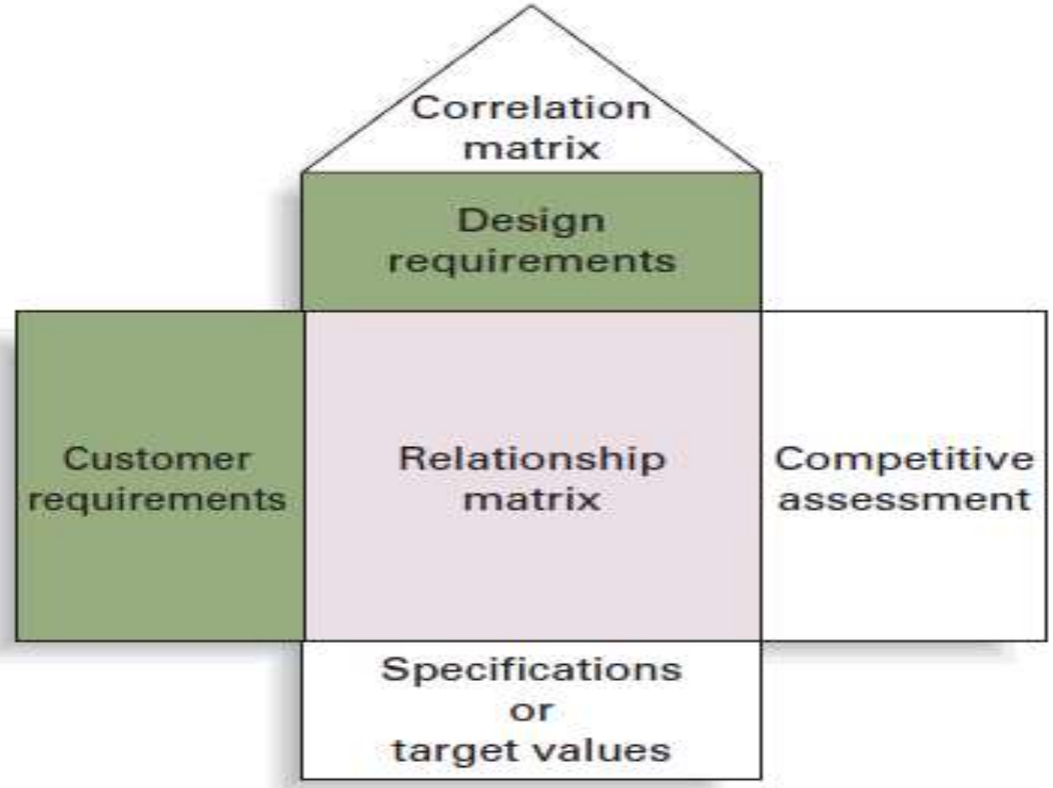
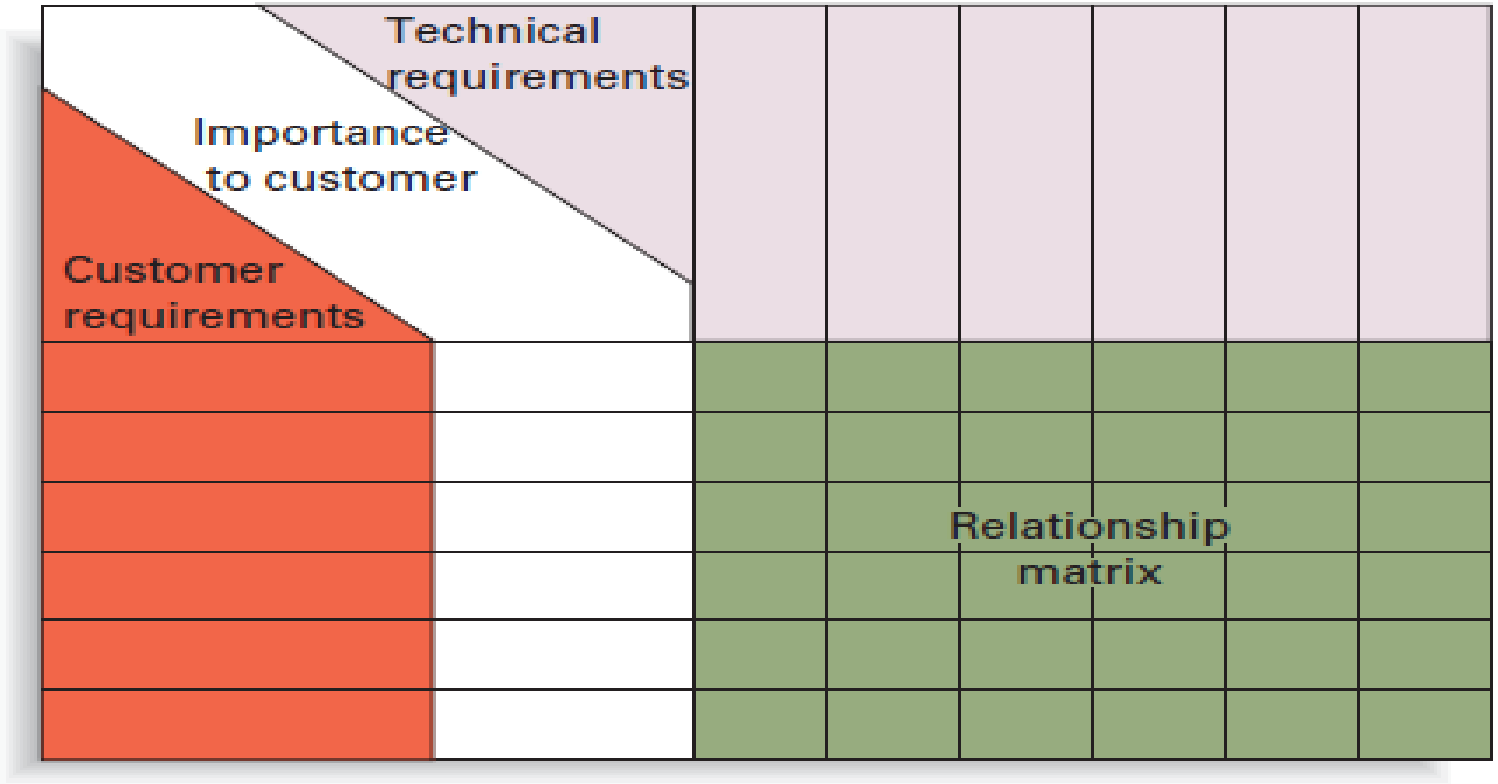


Source: Operation Management by William J Stevenson



Source: E& Y Consulting

QFD of Paper Business

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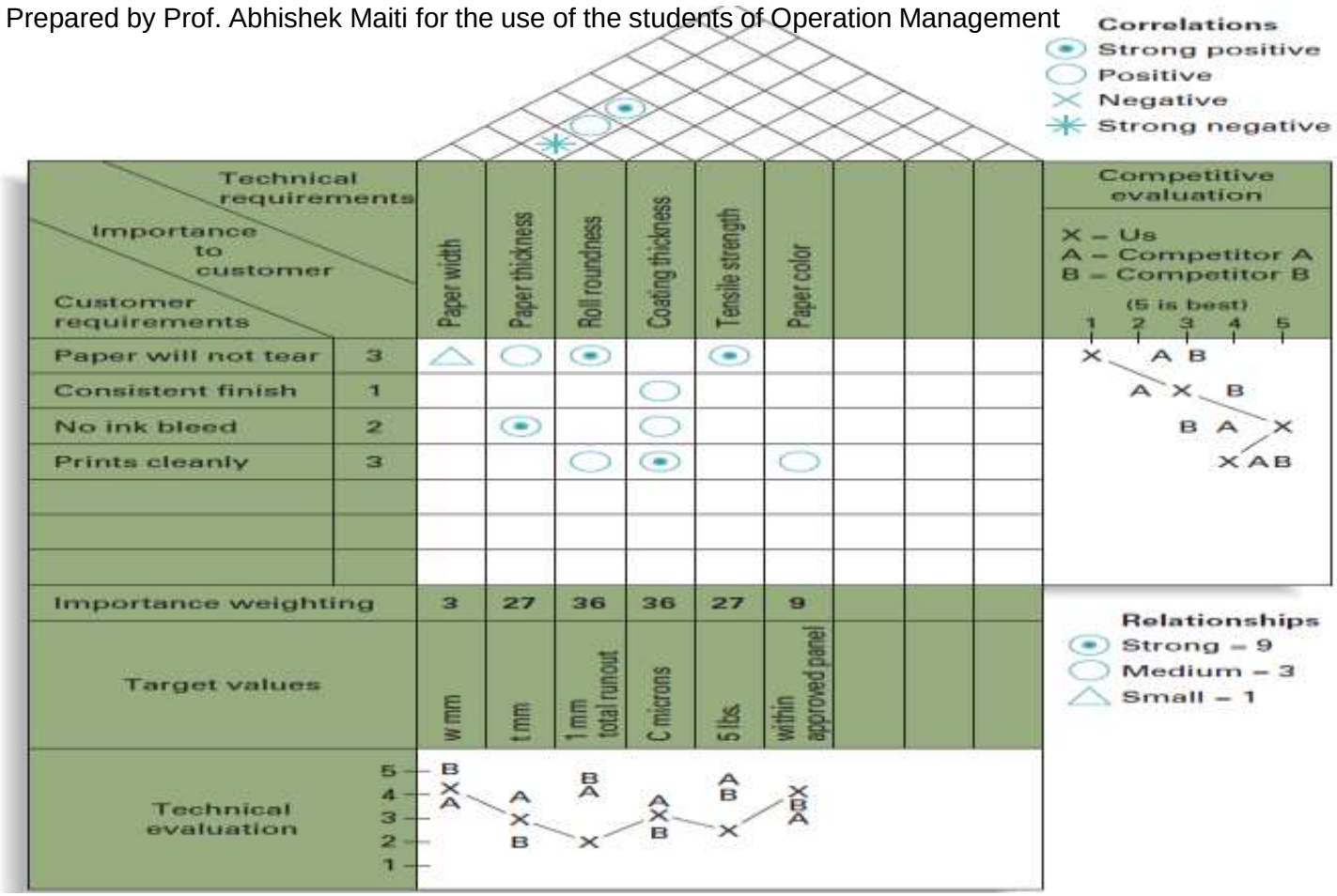

Source: Operation Management by William J Stevenson

Class Exercise

- Pizza Business.
- Let us list out basic customer wants.
- Give rating in 1-10 scale about the basic customer wants.
- Establish how to satisfy customer wants.
- Establish correlation between What and How using 1,3 and 9 scale.
- Evaluate.

Take Home Exercise

- Construct the House of Quality.

Organizing for Product Development

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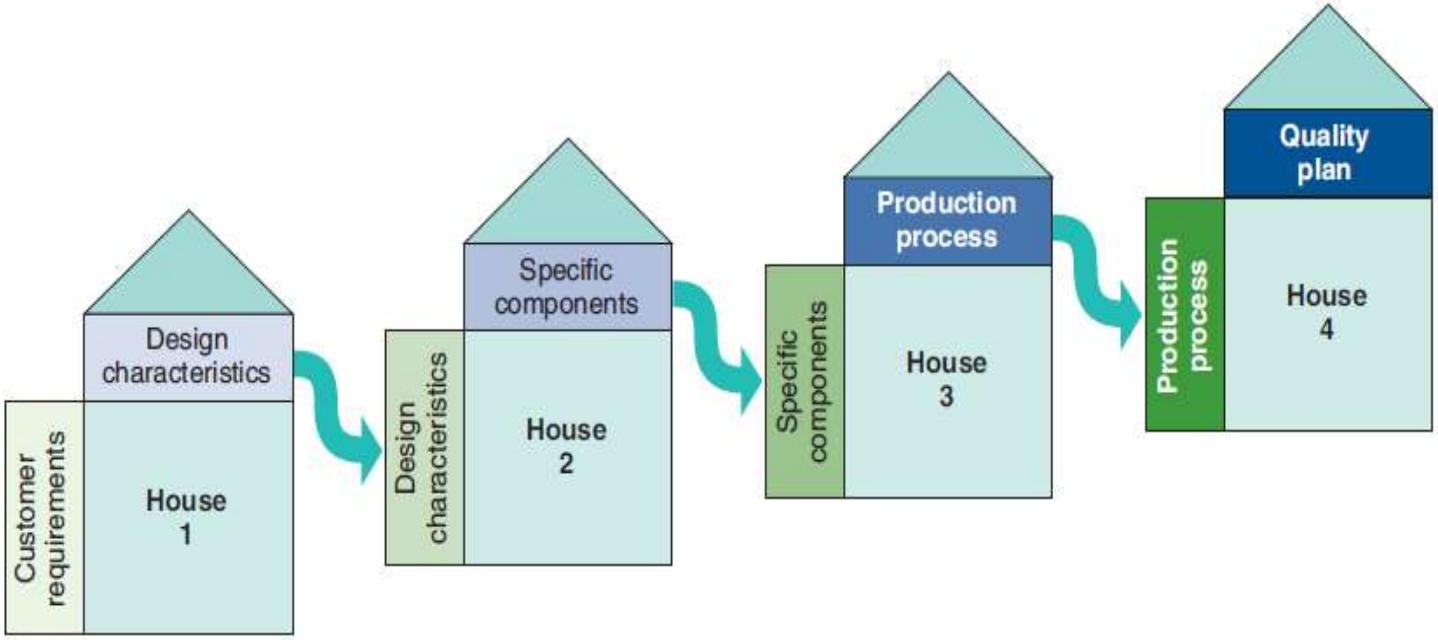

Figure 5.4

House of Quality Sequence Indicates How to Deploy Resources to Achieve Customer Requirements

Source: Heizer, Render, Muckson

Organizing for Product Development

There are four (4) approaches:

1. Traditional Approach.
2. Product Management Approach.
3. Team Approach.
4. Japanese Approach.

Manufacturability & Value Engineering

- Activities that help to improve a product's design, production, maintainability, and use. It helps to

1. Reduced complexity of the product.
2. Reduction of environmental impact.
3. Additional standardization of components.
4. Improvement of functional aspects of the product.
5. Improved job design and job safety.
6. Improved maintainability (serviceability) of the product.
7. Robust design.

Value Engineering

- Value Engineering is a systematic method to improve the "value" of goods or products and services by using an examination of function.
- Value $= \frac{Performance}{Cost}$
- Value (max) $= \frac{Performance\ (max)}{Cost\ (min}$
- To maximize value, increase the performance at same cost or reduce the cost at same performance.

Value Engineering Cycle

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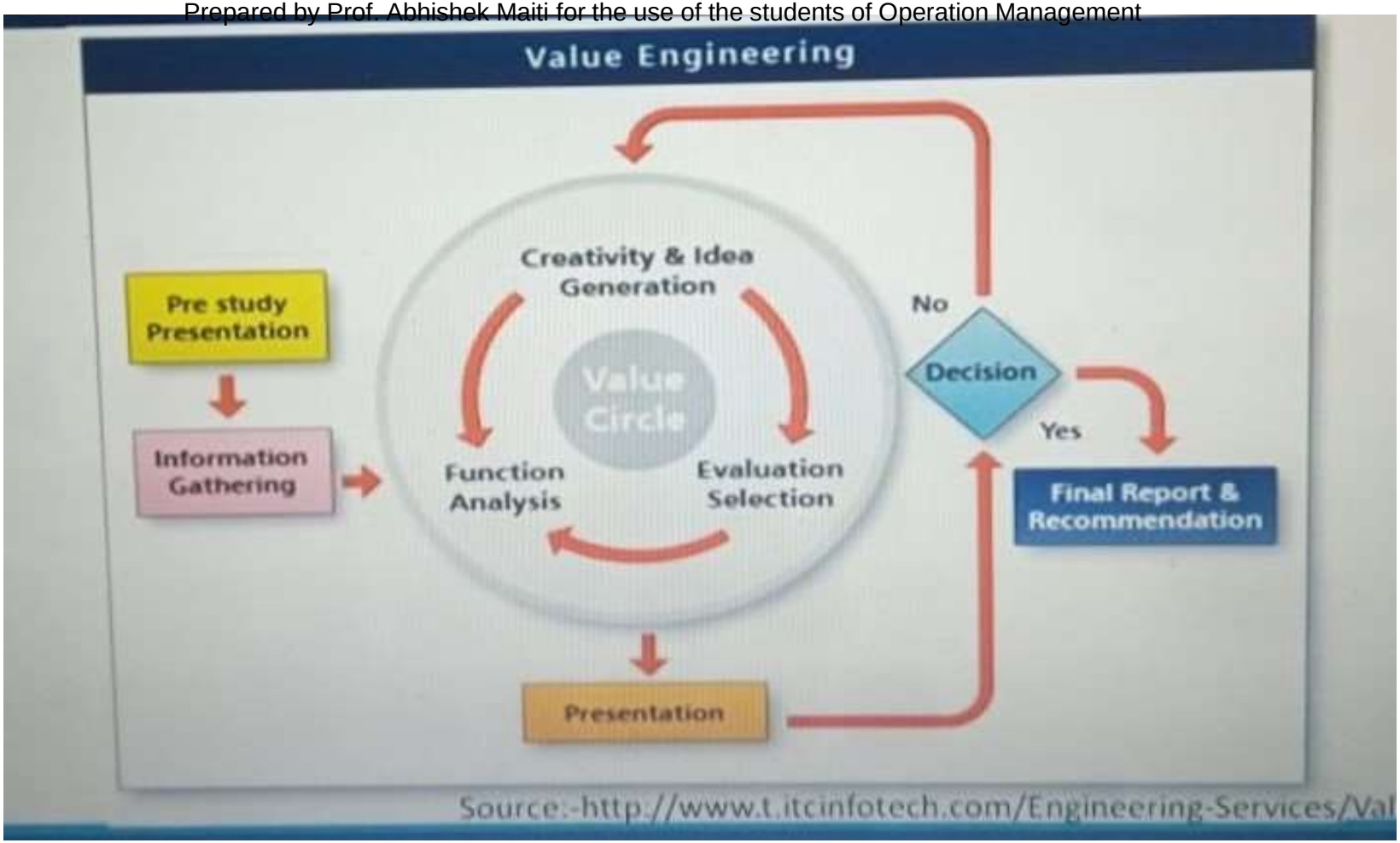
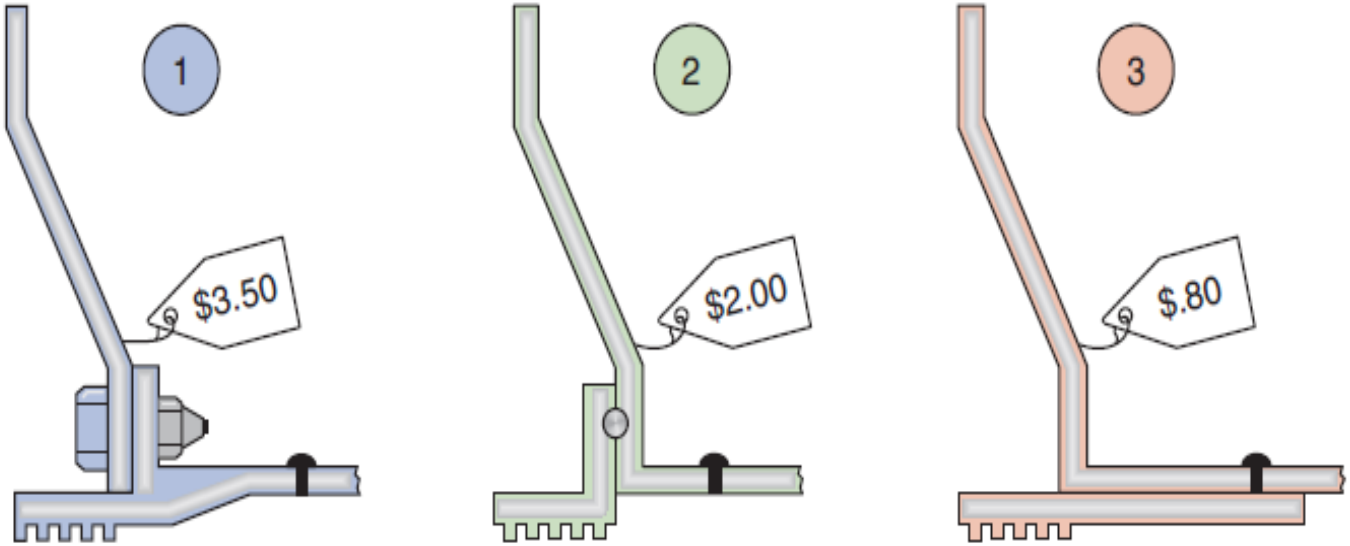


Value Engineering

Value engineering can be done by following ways:

- Specification : Identify non-essential elements.
- Design: Modify Design.
- Materials: Look for alternate materials.
- Process: Optimize Process

Class Discussion



Source: Heizer, Render, Muckson

Product Lifecycle (PLC)

- Product lifecycle shows the stages that a product go through development to decline from the market.
- It helps the firm to strategize about the NPD or manufacturing of the product.
- It helps the firm to plan when the production can be increased, reduced or stopped.
- It helps in forecasting, cash flow management and earning revenue.

Phases of PLC

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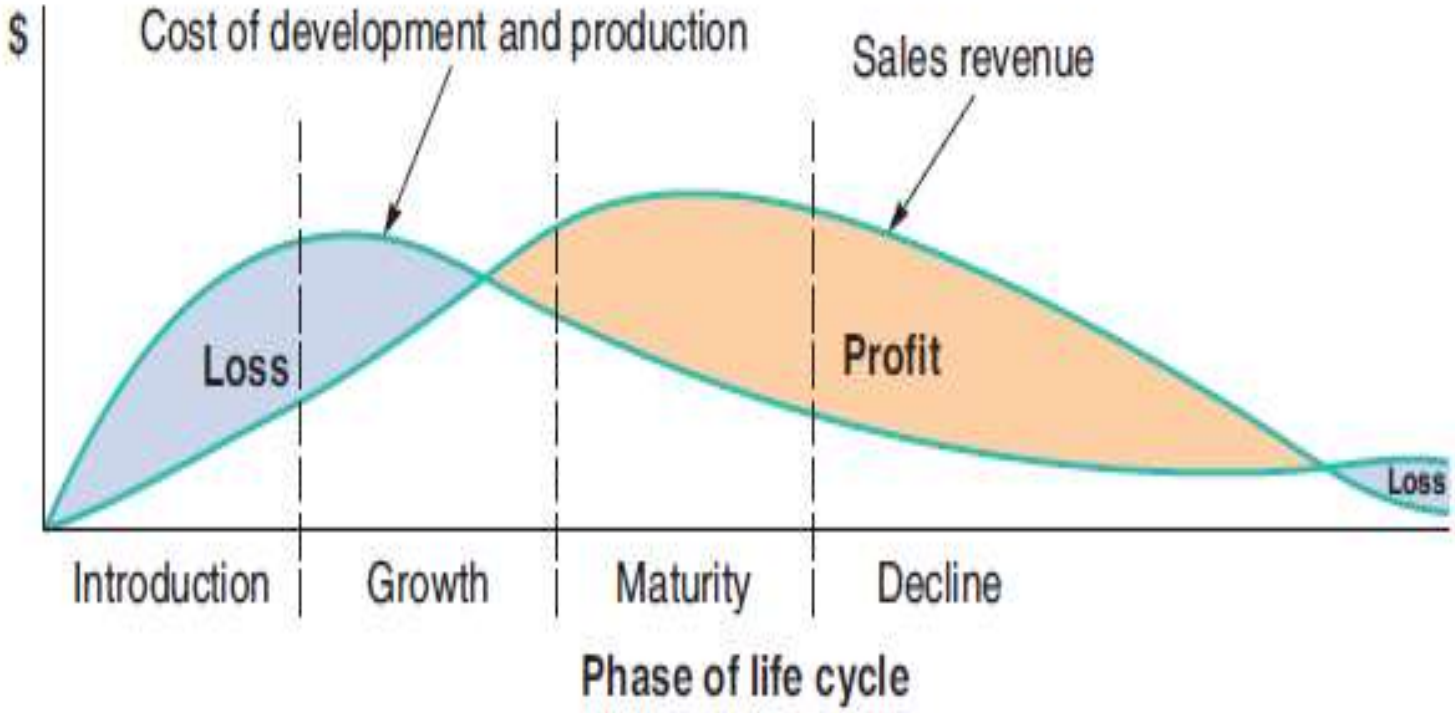

Source: Heizer, Render, Muckson

Phase 1: Introduction / Launch

- Introduction of old product into new market.
- Introduction of new product into the old market.
- Introduction of the new product into the new market.
- Supply > Demand
- Low sales.
- Advertisement and promotion.

Phase 1: Introduction / Launch

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Objectives	Characteristics
Sales	Low
Cost	High cost per customer
Competition	None
Profits	Negligible or Loss
Strategy	Market establishment, manufacturing at trial basis

Phase 2: Growth

- Demand increases, manufacture product.
- Optimizing the manufacturing strategy, pricing and discount policies.
- Revenue increases.

Phase 2: Growth

Objectives	Characteristics
Sales	Increases
Cost	Average cost per customer
Competition	Some
Profits	Positive
Strategy	Market penetration

Phase 3: Maturity

- Period of highest return.
- Supply < Demand.
- Manufacturing strategy dominates the marketing strategy.
- Manufacturing through mass production or economies of scale.
- Threat of new entrants.
- Bargaining power of suppliers.

Phase 3: Maturity

Objectives	Characteristics
Sales	Highest
Cost	Least cost per customer
Competition	Many rivals
Profits	High
Strategy	Maximize profit

Phase 4: Decline

- Competition in the market and rivalry among existing competitors.
- Manufacturing of the same product takes new route.
- From traditional manufacturing to FMS or JIT manufacturing.
- From mass production to mass customization.

Phase 4: Decline

Objectives	Characteristics
Sales	Declining
Cost	Marketing cost dominates
Competition	Few
Profits	Low
Strategy	Withdrawal or Alliance

Any Question

Issues for Product Design

- Robust Design.
- Customization.
- Computer-Aided Design (CAD) / Computer Aided Manufacturing (CAM).
- Virtual Reality Technology.
- Concurrent Engineering.
- Sustainability.

Robust Design

- Design that results in products or services that can function over a broad range of conditions.
- Do not adversely affected by small variations.

Customization

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Delayed differentiation

Modular design

CAD / CAM

- Computer-aided design (CAD) : Interactive use of a computer to develop and document a product.
- Computer-aided manufacturing (CAM) : The use of information technology to control machinery.
- Design for manufacture and assembly (DFMA) : Software that allows designers to look at the effect of design on manufacturing of the product.
- 3-D printing :An extension of CAD that builds prototypes and small lots.

Virtual Realty Technology

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Concurrent Engineering

- **Concurrent engineering :** Bringing engineering design and manufacturing personnel together early in the design phase.

Sustainability

- **Cradle-to-grave assessment :** The assessment of the environmental impact of a product or service throughout its useful life.
- End-of-life (EOL) programs deal with products that have reached the end of their useful lives.
- The 3 R of Sustainable Manufacturing : Reduce, Reuse and Recycle.

Sustainability

- **Reduce (Value analysis):** Examination of the function of parts and materials in an effort to reduce cost and/or improve product performance.
- **Reuse (Remanufacturing):** Refurbishing used products by replacing worn-out or defective components.
- **Recycling:** Recovering materials for future use.

Class Discussion

- Is 3 R of Sustainable manufacturing (Reduce, Reuse and Recycle) is applicable in your organization.
- Elaborate with few examples.

Product Development Continuum

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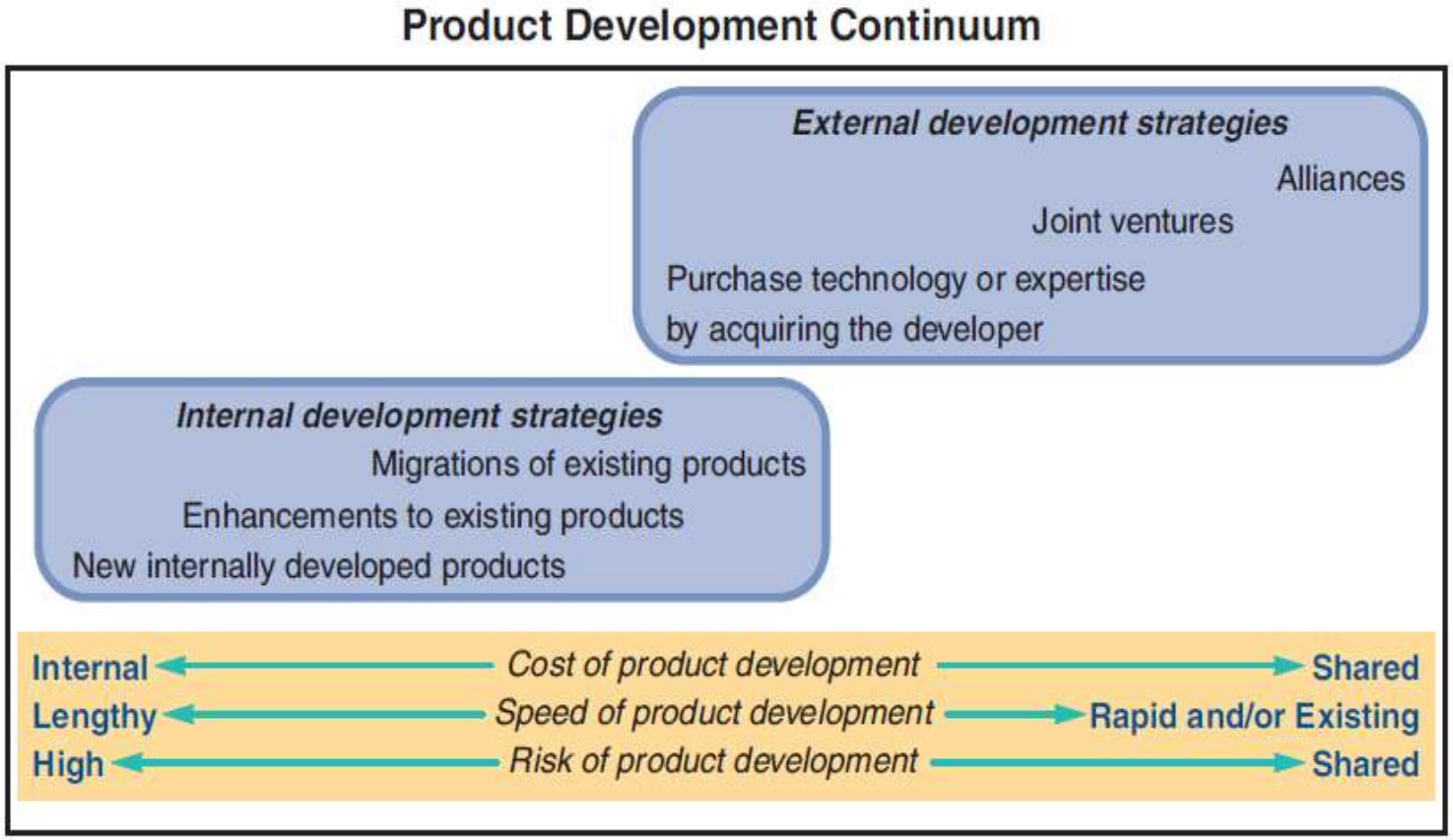

Source: Heizer, Render, Muckson

Any Question

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Design of Services

Service Design

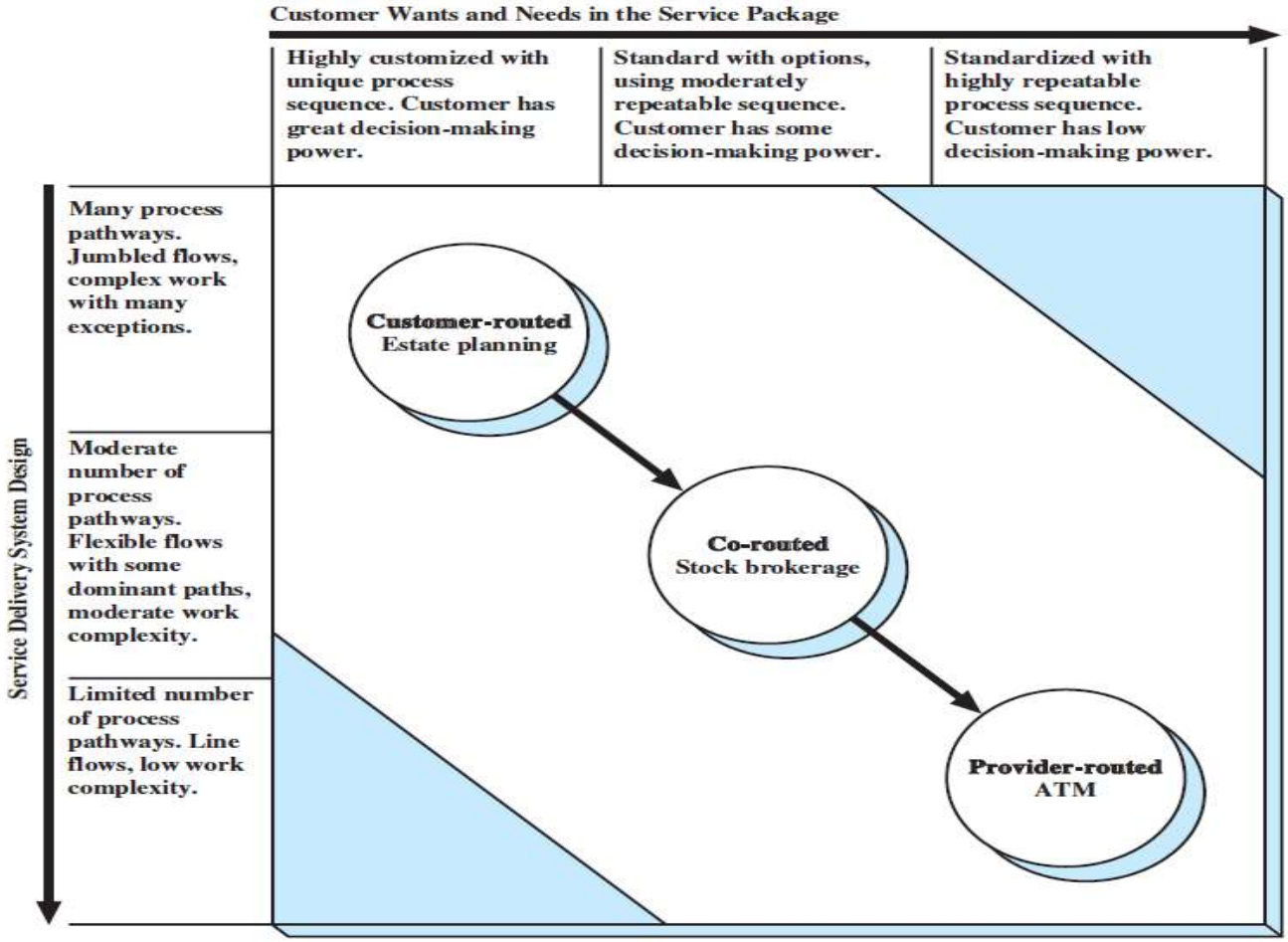
- Service.
- Service delivery system.
- Product bundle.
- Service package.
- Service blueprint.
- Process–chain–network (PCN) analysis.
- Process chain.

Phases in Service Design Process

- Conceptualize : Idea generation, Assessment of customer wants/needs (marketing), Assessment of demand potential (marketing)
- Identify service package components needed (operations and marketing).
- Determine performance specifications (operations and marketing).
- Translate performance specifications into design specifications.
- Translate design specifications into delivery specifications.

Service Delivery System Matrix

Source: Adapted from Collier and Meyer (1998).



Service-Profit Chain

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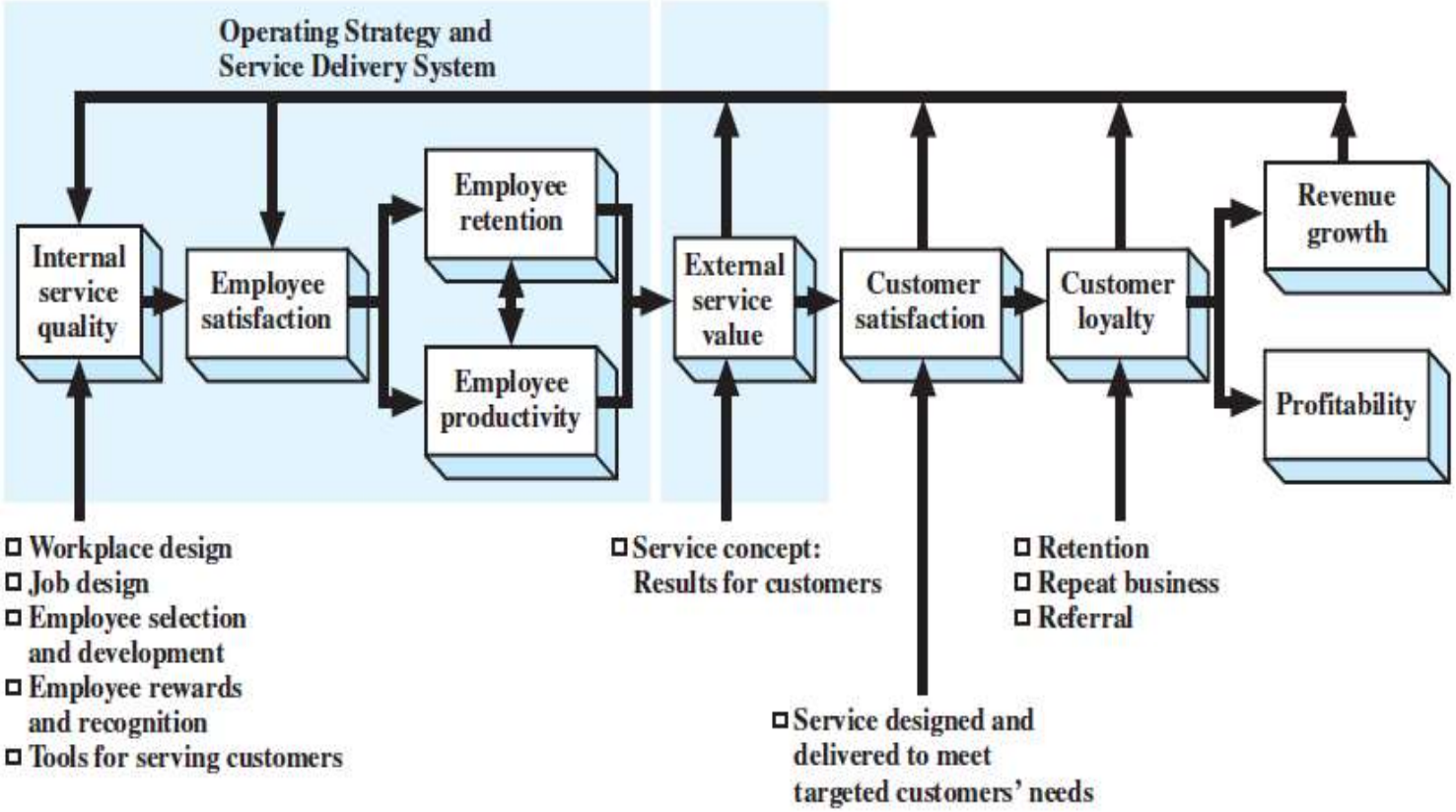
Source: Heskett et al., "Putting the Service-Profit Chain to Work," *Harvard Business Review*, March–April 1994, p. 166.



How to Improve Service Efficiency

- Limit the Options.
- Delay Customization.
- Modularization.
- Automation.
- Moment of Truth.

Take Home Exercise

- Identify the Similarity and Differences between Value Chain (Lecture 2 notes) and Service Profit Chain (Lecture 3 notes).

Further Reading / References

1. **Operation Management by Jay Heizer, Barry Render & Chuck Munson—Chapter 5**
2. Operation Management by William J Stevenson—Chapter 4
3. Operations Management in the Supply Chain by Roger G Schroeder, Susan Meyer Goldstein and M Johnny Rungtusanatham—Chapter 3 & 5

THANK YOU